

SECURED

Product Requirements Document

Pay rent. Earn cashback. Protect your landlord.

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Table of Contents

1. Executive Summary	3
2. Product Overview	4
3. User Flows	5
3.1 Authentication & OTP	5
3.2 Agreement Upload & Waitlist	7
3.3 Landlord Onboarding	10
3.4 Payment Flow	13
3.5 Aadhaar KYC	16
3.6 Referral System	18
4. Session Management	20
5. Edge Cases Matrix	23
6. Screen Inventory	26
7. Tech Stack	28
8. Database Schema	29

1. Executive Summary

Secured is a rent payment app that rewards tenants with **1% cashback** on rent payments while providing landlords with **vacancy protection** worth up to INR 1.5 lakhs.

Value Proposition

Stakeholder	Benefit
Tenants	1% cashback on rent, adjusted against next month's payment
Landlords	Vacancy assurance: 60-day replacement guarantee or 1 month rent coverage
Secured	Transaction fees, float income, tenant placement fees

Key Business Rules

- Vacancy cover activates after 3 successful rent payments
- Payment window: 1st - 7th of each month only
- One payment per month maximum
- Aadhaar KYC required before first cashback adjustment (Month 2)
- One active tenant per property (flatmate feature in V2)
- Name matching threshold: 80% for Indian name variations
- Max 10 referrals per user per month

2. Product Overview

User Journey Overview

Phase	Description	Duration
1. Onboarding	Phone OTP verification + Mobile 360 data fetch	~2 min
2. Waitlist	Agreement upload, OCR parsing, data confirmation	~5 min
3. Verification	Backend: SHCIL stamp check, risk scoring, coverage check	24-48 hrs
4. Landlord Setup	Electricity verification, bank penny drop, landlord confirmation	~10 min + wait
5. Active	Monthly rent payments, cashback accrual, KYC completion	Ongoing
6. Protected	Vacancy cover active after 3 payments	From Month 4

App Structure

Screen	Purpose	Access
Home	Pay rent, view status, quick actions	Post-onboarding
Transactions	Payment history, receipts, status tracking	Post-first payment
Profile	Personal info, house details, settings	Always
Referrals	Invite friends, track rewards	Post-approval

3. User Flows

3.1 Authentication & OTP Flow

Users authenticate via phone number with OTP delivered through SMS or WhatsApp. Background Mobile 360 API call fetches KYC data for pre-filling.

Flow States

State	Description	Next Action
SPLASH	App launch, 2 second display	Check auth status
PHONE_ENTRY	User enters 10-digit mobile	Validate & send OTP
CHANNEL_SELECT	User chooses SMS or WhatsApp	Send OTP
OTP_SENT	Waiting for user to enter OTP	Verify OTP
OTP_VERIFIED	OTP correct, fetch Mobile 360	Route to appropriate screen
LOCKED	3 wrong attempts, 15 min cooldown	Wait for timer

Edge Cases

#	Scenario	Handling
1	Invalid phone format	Inline validation: 'Enter valid 10-digit number'
2	SMS not delivered (60s)	Auto-show WhatsApp fallback option
3	OTP expired (2 min)	Clear input, show 'OTP expired. Resend?'
4	Wrong OTP (3x)	Lock for 15 min with countdown
5	App killed during OTP wait	Persist state, resume with remaining time
6	Mobile 360 API down	Silent fail, skip pre-fill, continue
7	Both SMS & WhatsApp fail	Show 'Contact support' option
8	Returning user (existing account)	Route to home/dashboard or resume flow

3.2 Agreement Upload & Waitlist Flow

Users upload their rental agreement (PDF or image) which is processed through OCR and Gemini AI for data extraction. Extracted data is confirmed by user before waitlisting.

Flow States

State	Description	Persisted Data
IDLE	Ready to upload	None
SELECTING	File picker open	None
VALIDATING	Checking file size/format	file_uri, file_size
UPLOADING	Chunked upload in progress	upload_id, chunks_completed, bytes_uploaded
PAUSED	Upload interrupted (recoverable)	resume_token, last_chunk_id
PARSING	Server OCR + Gemini extraction	server_file_id, ocr_job_id
CONFIRMING	User reviewing extracted data	parsed_data, user_edits
SUBMITTED	Waitlisted successfully	waitlist_position
FAILED	Upload or parsing failed	error_code, retry_count

Extracted Data Points (via Gemini)

Category	Fields
Tenant	name, phone (if present), aadhar_hint (last 4 digits)
Landlord	name, phone (if present), address
Property	full_address, city, state, pincode, property_type
Terms	monthly_rent, security_deposit, lease_start, lease_end, notice_period
Legal	stamp_paper_id, registration_number, execution_date

Edge Cases

#	Scenario	Handling
1	File too large (>10MB)	Error: 'File size exceeds 10MB'
2	Unsupported format	Error: 'Please upload PDF, JPG, or PNG'
3	Blurry/unreadable image	Error: 'Image quality too low' + tips
4	OCR extraction failure	Fallback: Manual entry form
5	Partial extraction	Show extracted fields, highlight missing

6	Multiple tenants in agreement	Flag primary, store all names
7	Agreement expired	Reject: 'Upload a current agreement'
8	App killed during upload (67%)	Resume modal: 'Resume from 67%?'
9	Network lost during upload	Pause, show 'Resume when online'
10	Duplicate property address	Block: 'Property already registered'

Backend Verification Pipeline

After waitlisting, backend runs automated verification checks to approve or reject users.

Check	Pass Criteria	Fail Action
SHCIL Stamp Verification	Valid stamp ID on SHCIL website	Flag for manual review
Property Deduplication	No other active tenant at address	Block with duplicate flow
Rent Range Check	INR 5,000 - 2,00,000/month	Reject if outside range
Geographic Coverage	Serviceable pincode	Reject: 'Expanding soon'
Lease Validity	>3 months remaining	Reject if <3 months left
Risk Score	Score above threshold	Route to manual review

3.3 Landlord Onboarding Flow

After user approval, they must verify their landlord through electricity bill ownership check, bank account penny drop, and landlord confirmation via email/WhatsApp.

Flow Steps

Step	Action	Verification
1. Intro	Explain landlord verification process	None
2. Electricity	Enter consumer number + provider	API: Name on bill vs agreement landlord
3. Ownership Match	Confirm if names match ($\geq 80\%$)	Fuzzy name matching algorithm
4. Bank Details	Enter landlord's bank account + IFSC	Penny drop API: Deposit INR 1
5. Bank Name Match	Confirm bank account holder name	Match $\geq 80\%$ with agreement
6. Contact	Enter landlord email/WhatsApp	None
7. Send Confirmation	Choose channel, send request	Email delivery / WhatsApp status
8. Pending	Wait for landlord response	Landlord clicks confirmation link

Name Matching Algorithm (Indian Names)

Indian names have many variations. The algorithm handles:

- Case differences: 'RAJESH' = 'Rajesh' (100%)
- Initial formats: 'R. Kumar' ~ 'Rajesh Kumar' (85%)
- Middle name: 'Rajesh Singh' ~ 'Rajesh Kumar Singh' (80%)
- Honorifics: 'Mr. Rajesh Kumar' = 'Rajesh Kumar' (100%)
- Spelling variations: 'Sharma' ~ 'Sharmaah' (90%)

Edge Cases

#	Scenario	Handling
1	Electricity bill in different name	'Bill in family member name?' + manual review
2	Penny drop fails (wrong account)	Clear, allow re-entry
3	Penny drop fails (bank downtime)	Retry with backoff, show 'Trying again...'
4	Bank name doesn't match ($< 80\%$)	Warning: 'Bank shows X. Agreement says Y.'
5	Joint bank account	Accept if ANY name matches $\geq 80\%$
6	NRI landlord (foreign bank)	Block: 'Indian accounts only'
7	Landlord email bounces	Notify user, request alternate
8	Landlord doesn't respond (7 days)	Auto-reminders at Day 3, 5; escalate at Day 7

9	Landlord rejects confirmation	Notify user: 'Contact landlord and retry'
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Landlord Confirmation (Web Page)

Landlords receive an email/WhatsApp with a link to confirm their tenant. This is a lightweight web page (not in the app).

#	Scenario	Handling
1	Link expired (>7 days)	Show: 'Link expired. Ask tenant to resend.'
2	Link already used	Show: 'Already confirmed.'
3	Wrong person clicks	Allow rejection: 'Not your tenant?'
4	Landlord confirms	Show success + vacancy cover info
5	Landlord rejects	Notify tenant, allow new landlord contact

3.4 Payment Flow

Users can pay rent through the app between the 1st and 7th of each month. Cashback (1%) is earned on each payment and applied to next month's rent.

Payment Window Rules

Date Range	Button State	User Experience
Before 1st	Disabled + Timer	'Pay Rent - Opens in 3 days'
1st - 7th (not paid)	Enabled	'Pay Rent - January'
1st - 7th (paid)	Success state	'January Rent Paid' + receipt
8th - End (not paid)	Locked + Warning	'Window closed. Pay landlord directly.'

Payment State Machine

State	Description	Recovery Action if App Killed
IDLE	Ready to pay	None needed
CREATING_ORDER	Server order creation	Check if order exists
GATEWAY_OPENED	Razorpay/Cashfree webview	CRITICAL: Check server immediately
USER_PAID	User completed on gateway	Verify payment status
VERIFYING	Confirming with server	Retry verification
PROCESSING	Depositing to landlord	Show 'Processing' status
COMPLETED	Landlord received funds	Show success
FAILED	Payment failed	Show error + retry option
UNKNOWN	App killed mid-payment	Check server, show appropriate screen

Critical Edge Cases

#	Scenario	Handling
1	App killed while gateway open	CRITICAL: Check server on launch, show result
2	Payment done, app killed before verification	Check server -> show 'Payment found!'
3	Double payment attempt (same month)	Block: 'Already paid January rent'
4	Payment after window closes	Block, show 'Pay landlord directly'
5	Gateway timeout	Check server, show 'Verifying your payment...'
6	Network lost during verification	Retry with backoff, persist state

7	Payment succeeds, payout fails	Show 'Processing' + landlord payout status
8	Phone call during gateway	iOS preserves WebView; check status on return
9	First payment (no cashback yet)	Show: 'Cashback applied from next month'
10	KYC not done at Month 2	Block cashback: 'Complete KYC to unlock INR 250'

Payment Recovery Screens

Screen	When Shown	User Actions
'Checking Your Payment'	App launch with pending payment session	Wait (auto-check)
'Payment Found!'	Server confirms successful payment	[Done] button
'Payment Status Unknown'	Can't determine status	[Check Again] [Contact Support]
'Payment Failed'	Server confirms failure	[Retry] [View Details]

3.5 Aadhaar KYC Flow

KYC is triggered when user tries to apply cashback (Month 2). Verification happens via DigiLocker, India's official document wallet.

Flow States

State	Description	Next Step
KYC_PROMPT	Modal during payment: 'Unlock cashback'	User taps 'Complete KYC'
CONSENT	Explain data access, get consent	User agrees
DIGILOCKER_REDIRECT	Open DigiLocker in browser/webview	User authenticates
SUCCESS	Aadhaar data received	Name match check
NAME_MATCH_PASS	Name matches agreement ($\geq 80\%$)	KYC complete
NAME_MISMATCH	Name doesn't match ($< 80\%$)	Manual review request
FAILED	DigiLocker error or user cancelled	Retry prompt
SKIPPED	User skips KYC	No cashback, prompt again next month

Data Accessed from Aadhaar

Accessed: Full Name, Date of Birth, Photo

NOT Accessed: Aadhaar Number, Address, Other documents

Edge Cases

#	Scenario	Handling
1	User skips KYC	Allow payment, but no cashback adjustment
2	DigiLocker down	'DigiLocker unavailable. Try later.'
3	User cancels on DigiLocker	'KYC incomplete. Try again?'
4	App killed during DigiLocker	Check callback status on launch
5	Aadhaar name $\neq$ Agreement name	Show mismatch, offer manual review
6	Multiple KYC attempts ($> 3/\text{day}$)	'Too many attempts today'

3.6 Referral System

Active users can invite friends using their unique referral code. Rewards are unlocked when the referred user makes their first payment.

Reward Structure

Party	Reward	Trigger
Referrer	INR 500	Friend completes first payment
Referee (Friend)	INR 250	Completes first payment

Referral States

State	Description	Shown to Referrer
SIGNED_UP	Friend created account	'Amit signed up'
WAITLISTED	Friend uploaded agreement	'Amit is on waitlist'
APPROVED	Friend passed verification	'Amit approved'
PAID	Friend made first payment (REWARD!)	'Amit paid - +INR 500'
REJECTED	Friend didn't qualify	'Amit wasn't eligible'

Edge Cases

#	Scenario	Handling
1	Max referrals reached (10/month)	'Hit limit. Try again next month!'
2	Self-referral attempt	Block (device/phone fingerprint)
3	Invalid referral code	'Code not found. Check spelling.'
4	Friend rejected	No reward, show 'Didn't qualify'
5	Referrer churns before reward	Forfeit referrer reward; friend still gets bonus
6	Code entered after signup	Not allowed; must enter during signup
7	Payment refunded after reward	Claw back reward from balance

4. Session Management

Robust session management is critical for handling app kills, backgrounding, network drops, and crashes. Every critical flow must persist state to survive interruptions.

Session Protection Priority

Flow	Risk Level	Why Critical
Payment	CRITICAL	Money involved, gateway handoff, verification
Agreement Upload	High	Large file, multi-step parsing
Aadhaar KYC	Medium	External DigiLocker handoff
OTP Verification	Medium	Timer state, retry limits
Landlord Bank Verification	Medium	Penny drop API, async verification

Session Lifecycle Rules

Rule	Description
Create Early	Create session BEFORE any server call, generate UUID immediately
Persist Often	Update after EVERY state change, chunk upload, user input
Recover Smart	Payment: ALWAYS check server. Upload: Show resume prompt.
Expire Appropriately	Upload: 24h. Payment: Never (check server). OTP: 10 min.
Clean Up	Clear on completion, explicit cancel, or expiry. Never clear payment until confirmed.

App Launch Recovery Sequence

On every app launch, check for pending sessions in this priority order:

Priority	Check	Action
1. CRITICAL	Pending PAYMENT sessions	Block UI, verify with server immediately
2. HIGH	Pending KYC sessions	Check DigiLocker callback status
3. MEDIUM	Pending UPLOAD sessions	Show resume modal (non-blocking)
4. LOW	Pending OTP sessions	If valid (<2 min): resume. Else: clear.
5. NORMAL	User's current state	Navigate to appropriate screen

Upload Session - Persisted Data

State	Persisted Fields
SELECTED	file_uri, file_size, file_hash, file_bookmark
UPLOADING	upload_id, bytes_uploaded, total_chunks, completed_chunks[]
PAUSED	resume_token, last_chunk_id, error_reason
PARSING	server_file_id, ocr_job_id
CONFIRMING	parsed_data, user_edits, timestamp
FAILED	error_code, retry_count, last_attempt

Payment Session - Persisted Data

State	Persisted Fields
INITIATED	state_id, amount, month, created_at
CREATING_ORDER	Above + server request timestamp
GATEWAY_OPENED	order_id, gateway_order_id, gateway_type
USER_PAID	gateway_payment_id, gateway_signature
VERIFYING	verification_attempts, last_attempt
UNKNOWN	All above + last_known_status

5. Edge Cases Matrix

Complete matrix of edge cases across all flows, organized by category.

Network & Connectivity

Scenario	Affected Flows	Handling
Network lost mid-action	All	Pause, persist state, show offline indicator
Slow network (timeout)	Upload, Payment	Show progress, allow cancel, retry option
Network restored	All	Auto-resume for uploads; prompt for payments
Airplane mode toggle	All	Same as network lost/restored

App Lifecycle

Scenario	Affected Flows	Handling
App backgrounded	All	Save state, continue background tasks (30s iOS)
App killed (force quit)	All	Persist state survives; recover on launch
App crash	All	Same as kill; may have incomplete state
Phone call interruption	Payment, KYC	Same as backgrounded; check status on return
Low memory warning	Upload	Pause upload, save progress
App update required	All	Block app, show update prompt

Data & Validation

Scenario	Affected Flows	Handling
Duplicate phone number	Auth	Allow login (existing user flow)
Duplicate property address	Upload	Block: 'Property already registered'
Name mismatch (<80%)	Landlord, KYC	Show warning, allow override or manual review
Expired agreement	Upload	Reject: 'Upload current agreement'
Invalid bank account	Landlord	Penny drop fails; allow re-entry
Fraudulent document	Upload	Backend detection; manual review queue

External Service Failures

Service	Failure Scenario	Handling
Mobile 360 API	Timeout or error	Silent fail, skip pre-fill
OCR Service	Can't read document	Fallback to manual entry
Gemini AI	Parsing failed	Partial data + manual entry
SHCIL Website	Scraping blocked	Flag for manual verification
Electricity API	Provider not supported	'Other' option, manual entry
Penny Drop API	Bank downtime	Retry with backoff, show progress
DigiLocker	Service unavailable	'Try again later' with retry
Payment Gateway	Timeout	Check server for actual status
Email Service	Bounce	Notify user, request alternate
Push Notifications	Delivery failed	Fallback to SMS for critical alerts

Business Rule Edge Cases

Scenario	Rule	Handling
Payment outside window	1st-7th only	Disable button: 'Window closed'
Second payment same month	One per month	Block: 'Already paid January'
Rent amount changed	Matches agreement	Require new agreement upload
Lease ending soon	<3 months	Proactive prompt: 'Lease ends in X days'
User wants to move out	Vacancy claim trigger	Notice flow, 60-day clock starts
Flatmate tries to sign up	One tenant per property	Block: 'Flatmate feature coming soon'
Max referrals reached	10 per month	'Limit reached. Try next month!'

6. Screen Inventory

Complete list of screens to be designed and developed.

Flow	Screen	Type
Auth	Splash	Full screen
Auth	Phone Entry	Full screen
Auth	OTP Channel Selection	Full screen
Auth	OTP Entry	Full screen
Auth	Locked (Max Attempts)	Full screen
Upload	Agreement Upload	Full screen
Upload	Upload Progress	Full screen
Upload	Parsing Progress	Full screen
Upload	Data Confirmation	Full screen
Upload	Waitlist Confirmation	Full screen
Upload	Upload Failed	Full screen
Upload	Resume Upload Modal	Modal
Upload	Network Lost Overlay	Overlay
Landlord	Verification Intro	Full screen
Landlord	Electricity Details	Full screen
Landlord	Ownership Match Warning	Full screen
Landlord	Bank Details	Full screen
Landlord	Bank Name Match Warning	Full screen
Landlord	Landlord Contact	Full screen
Landlord	Pending Confirmation	Full screen
Landlord (Web)	Confirmation Page	Web page
Landlord (Web)	Confirmed + Vacancy Info	Web page
Payment	Home (Pay Rent)	Full screen
Payment	Payment Summary	Full screen
Payment	Processing	Full screen
Payment	Success	Full screen

Flow	Screen	Type
Payment	Failed	Full screen
Payment	Verifying (Recovery)	Full screen
Payment	Payment Found (Recovery)	Full screen
Payment	Payment Unknown	Full screen
KYC	KYC Prompt (in Payment)	Modal
KYC	Consent	Full screen
KYC	Success	Full screen
KYC	Name Mismatch	Full screen
Referral	Referral Home	Full screen
Referral	Enter Code (Signup)	Full screen
Transactions	Transaction List	Full screen
Transactions	Transaction Detail	Full screen
Transactions	Zero State	Full screen
Profile	Profile Overview	Full screen
Profile	Personal Info Edit	Full screen
Profile	House Info	Full screen
Profile	Bank Account Edit	Full screen
Profile	Notification Settings	Full screen
Profile	Delete Account Confirm	Modal
Move-out	Confirm Intent	Full screen
Move-out	Date Selection	Full screen
Move-out	Notice Sent	Full screen
Global	No Connection	Overlay
Global	Session Expired	Modal
Global	Force Update	Full screen

Total Screens: ~51

7. Tech Stack

Recommended iOS Stack

Layer	Technology	Notes
UI Framework	SwiftUI + UIKit	SwiftUI primary; UIKit for complex custom components
Architecture	MVVM + Clean Architecture	With Swift Concurrency (async/await)
State	Combine + @Observable	Reactive streams for real-time updates
Database	Supabase (PostgreSQL)	Hosted, real-time, auth included
Auth	Supabase Auth + Custom OTP	OTP via MSG91/Gupshup
Storage	Supabase Storage	For agreement documents
Local Storage	MMKV	Crash-safe session persistence
Networking	URLSession / Alamofire	Background URLSession for uploads
Images	Kingfisher	Caching and loading
Animations	Lottie	Micro-animations (cashback celebration)
Security	Keychain + CryptoKit	Secure storage, encryption
Biometrics	LocalAuthentication	Face ID / Touch ID
Push	Firebase Cloud Messaging	Notifications
Analytics	Mixpanel / Amplitude	User behavior tracking
Crash Reporting	Sentry	Error monitoring

Third-Party Integrations

Service	Provider	Purpose
OTP (SMS)	MSG91 / Gupshup	Phone verification
OTP (WhatsApp)	Gupshup / Twilio	Phone verification
Mobile 360	Cashfree Secure ID	KYC data pre-fill
OCR	Google Cloud Vision	Agreement text extraction
AI Parsing	Google Gemini	Structured data extraction
Penny Drop	Cashfree / Razorpay	Bank account verification
Aadhaar KYC	DigiLocker / Cashfree OKYC	Identity verification

Payments	Razorpay / Cashfree	Rent collection
Payouts	Razorpay / Cashfree	Landlord disbursement
Email	Resend / SendGrid	Landlord confirmation
Web Scraping	Firecrawl / Hyperbrowser	SHCIL stamp verification

8. Database Schema (Supabase)

Core Tables

Table	Purpose	Key Fields
users	User accounts	id, phone, email, status, aadhaar_verified, referral_code
properties	Rental properties	id, full_address, city, pincode, normalized_address
landlords	Landlord info	id, name, bank_account, bank_verified, confirmed_at
rental_agreements	Uploaded agreements	id, user_id, property_id, monthly_rent, parsed_data
tenancies	User-Property link	id, user_id, property_id, vacancy_cover_status
payments	Rent payments	id, tenancy_id, amount, status, gateway_payment_id
cashback_ledger	Cashback transactions	id, user_id, type, amount, balance_after
referrals	Referral tracking	id, referrer_id, referee_id, status, reward_paid
verifications	Audit trail	id, user_id, type, status, match_score
vacancy_claims	Landlord claims	id, tenancy_id, triggered_at, resolution_type
notifications	Push/Email log	id, user_id, type, channel, sent_at

Key Enums

Enum	Values
user_status	pending_agreement, waitlisted, under_review, approved, rejected, active, churned
payment_status	initiated, processing, completed, depositing, deposited, failed, refunded
vacancy_cover_status	inactive, building (1-2 payments), active (3+), claimed, expired
referral_status	signed_up, waitlisted, approved, paid, rejected
verification_status	pending, in_progress, passed, failed, manual_review